Alessio **Novel**

Artificial Intelligence Student

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Education

Università degli Studi di Trento — *Trento, Italy*

M.Sc. in Artificial Intelligence Systems (Code: LM-32) – Neuroscience Track — *Sep 2024 – Present*

- **Current GPA**: 28.8/30
- **Key Courses**: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Foundations of Cognitive Psychology and Neuroscience, Neuroimaging for Data Science

Università degli Studi di Trento — *Trento, Italy*

B.Sc. in Computer Science (Code: L-31) — *Sep 2021 – Jul 2024*

- **Graduation Grade**: 110/110 with honors
- **Key Courses**: Algorithms and Data Structures, Databases, Networking, Computer and Network Security, Machine Learning, Human-Computer Interaction, Information Systems

Istituto Tecnico Statale A. Volta — *Trieste, Italy*

Secondary School Diploma in computer science (Code: D040) — *Sep 2016 – Jun 2021*

- **Graduation Grade**: 100/100

Work Experience

CAIAN (Centre for AI and Neuroscience) - University of Bonn — *Bonn, Germany*

AI & Neuroscience Research Intern — *Aug 2026 – Present*

- Conducting research on **representational alignment** within a global multi-lab collaboration collecting the largest single-trial fMRI dataset on **fear conditioning**, supervised by Prof. Dr. Dr. Dominik Bach and Dr. Jurjen Heij.
- Preprocessing fMRI brain imaging data and building geometric analysis pipelines to evaluate human neural representations of fear responses.
- Designing and training brain-inspired Recurrent Neural Networks with Reinforcement Learning to model fear conditioning learning and compare artificial vs. biological neural representations.

Projects

Few-Shot CLIP Adaptation (Flowers102 dataset)

GitHub Repository

- Developed and compared new few-shot learning strategies (based on CoOp) to adapt the CLIP vision-language model for fine-grained classification on Oxford Flowers102 dataset.
- The goal was to improve accuracy using only 10 samples for half of the classes, while not having samples for the other half. Improved Base Accuracy by **11.49 percentage points** (from 66.92% to 78.41%) and increased the overall Harmonic Mean by **5.75 percentage points** compared to the zero-shot baseline, achieving the objective of improving base class accuracy while minimizing degradation on novel class performance.

Skills

Programming	Python, C++, Java, R
Back-end	PHP, REST API, MySQL, PostgreSQL, MongoDB
Front-end	HTML5, CSS3, JavaScript
DevOps	Git, Conda, Docker, CMake
Languages	Italian (native), English

Extracurricular Activities

M2L Summer School — *Milan, Italy*

Participant — *Sep 2024*

- Selected among 519 applicants (34% acceptance rate) for an immersive Machine Learning summer school. Engaged with industry experts on advanced AI topics through intensive technical lectures, workshops, and networking sessions.

Olympic Winter Games Milano Cortina 2026 — *Tesero, Italy*

Technology Volunteer — *Jul 2025 – Feb 2026*

- After specialized training, I managed IT setup and connectivity for media and staff at the Cross-Country Skiing venue, overseeing device distribution and network support.